

Tutorial Solutions

Tutorial 01: Population, Sample & Descriptive Statistics

Population vs sample, parameter vs statistic, Measures of central tendency, Measures of dispersion

Learning outcomes:

- ✚ Differentiate between population, sample, parameter, and statistic.
- ✚ Apply measures of central tendency and dispersion to summarize data appropriately.

1. A data scientist records the daily sales (in units) of a product for 10 days:

45, 50, 52, 48, 55, 60, 62, 58, 50, 70

a) Calculate the mean, median, and mode.

b) Determine the range and standard deviation.

Solution:

Given data (daily sales in units):

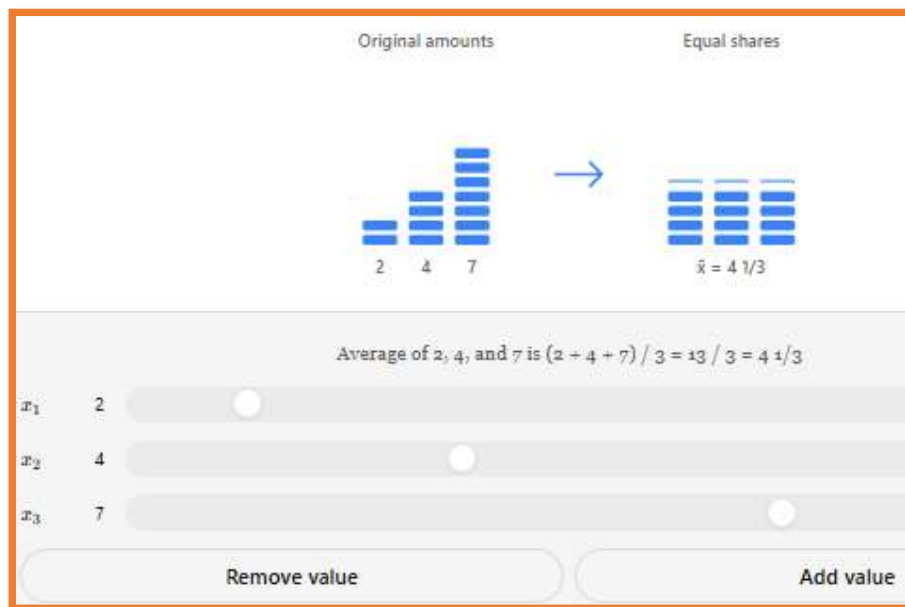
45, 50, 52, 48, 55, 60, 62, 58, 50, 70

(a) Mean, Median, and Mode

Step 1: Arrange the data in ascending order

45, 48, 50, 50, 52, 55, 58, 60, 62, 70

Mean



$$\text{Mean} = \frac{\sum x}{n}$$

Sum of all observations:

$$45 + 50 + 52 + 48 + 55 + 60 + 62 + 58 + 50 + 70 = 550$$

Number of observations:

$$n = 10$$

$$\text{Mean} = \frac{550}{10} = 55$$

Median

Since there are **10 observations (even number)**,

$$\text{Median} = \frac{\text{5th value} + \text{6th value}}{2}$$

$$= \frac{52 + 55}{2} = \frac{107}{2} = 53.5$$

$$\text{Median} = 53.5$$

Mode

The value occurring most frequently is:

- a) 50 appears **2 times**
- b) All other values appear once.

$$\text{Mode} = 50$$

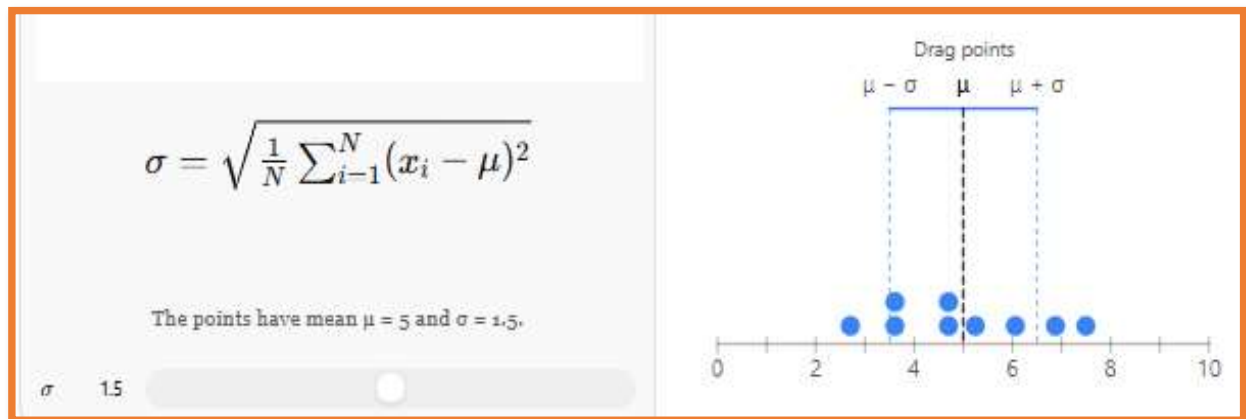
(b) Range and Standard Deviation

Range

$$\begin{aligned}\text{Range} &= \text{Maximum} - \text{Minimum} \\ &= 70 - 45 = 25\end{aligned}$$

$$\text{Range} = 25$$

Standard Deviation



Using the **population standard deviation** formula:

x	$x - 55$	$(x - 55)^2$
45	-10	100
50	-5	25
52	-3	9
48	-7	49
55	0	0
60	5	25
62	7	49
58	3	9
50	-5	25
70	15	225
Total		516

Variance:

$$\sigma^2 = \frac{516}{10} = 51.6$$

Standard deviation:

$$\sigma = \sqrt{51.6} = 7.18$$

Population Standard Deviation ≈ 7.18
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If this data is treated as a **sample**:

$$s = \sqrt{\frac{516}{9}} = \sqrt{57.33} \approx 7.57$$

Sample Standard Deviation ≈ 7.57
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2. A company has 5,000 employees. A researcher selects 100 employees and records their monthly salaries.

a) Distinguish between population and sample in this context.

b) Distinguish between a parameter and a statistic.

c) The salaries (₹ in thousands) of 10 sampled employees are:

25, 28, 30, 35, 40, 42, 45, 50, 55, 60

Calculate the mean salary and variance.

Solution:

(a) Distinguish between Population and Sample

Population	Sample
The entire group of individuals under study.	A subset of the population selected for analysis.
Consists of all 5,000 employees in the company.	Consists of the 100 selected employees .
Usually large and difficult to study completely.	Smaller, easier, and less expensive to study.
Provides complete information about the company.	Used to make inferences about the population.

Answer:

- a) **Population:** All **5,000 employees** of the company.
- b) **Sample:** The **100 employees** selected by the researcher.

(b) Distinguish between a Parameter and a Statistic

Parameter	Statistic
A numerical measure describing a population .	A numerical measure describing a sample .
Usually denoted by symbols such as μ (mean) or σ^2 (variance).	Usually denoted by $\bar{x}$ (sample mean) or s^2 (sample variance).
Fixed but generally unknown.	Calculated from sample data and varies from sample to sample.

Answer:

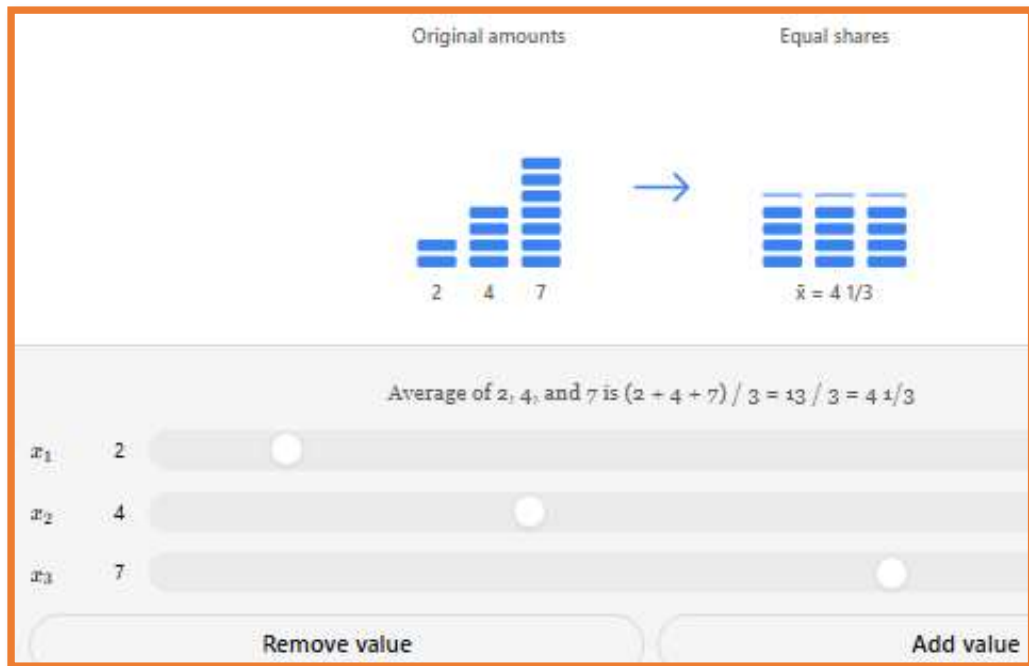
- a) **Parameter:** The actual average salary of all **5,000 employees**.
- b) **Statistic:** The average salary calculated from the **100 sampled employees**.

(c) Calculate the Mean Salary and Variance

Given salaries (₹ in thousands)

25, 28, 30, 35, 40, 42, 45, 50, 55, 60

Step 1: Calculate the Mean



$$\bar{x} = \frac{\sum x}{n}$$

Sum of salaries:

$$25 + 28 + 30 + 35 + 40 + 42 + 45 + 50 + 55 + 60 = 410$$

Number of observations:

$$n = 10$$

$$\bar{x} = \frac{410}{10} = 41$$

Mean Salary = ₹41 thousand

Step 2: Calculate the Variance



Salary (x)	$x - 41$	$(x - 41)^2$
25	-16	256
28	-13	169
30	-11	121
35	-6	36
40	-1	1
42	1	1
45	4	16
50	9	81
55	14	196
60	19	361
Total		1238

Sample Variance

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$$

$$= \frac{1238}{9} = 137.56$$

$$\text{Sample Variance} = 137.56 \text{ (thousand rupees)}^2$$

(Optional) Population Variance

If these 10 salaries are treated as the entire population,

$$\sigma^2 = \frac{1238}{10} = 123.8$$

$$\text{Population Variance} = 123.8 \text{ (thousand rupees)}^2$$

3. The response times (in milliseconds) of a web server are: 120, 130, 125, 140, 135, 128, 500

a) Calculate the mean and median.

b) Which measure of central tendency is more appropriate for this dataset? Justify your answer.

c) Calculate the range and interquartile range (IQR).

Solution:

(a) Calculate the Mean and Median

Step 1: Arrange the data in ascending order

120, 125, 128, 130, 135, 140, 500

Mean

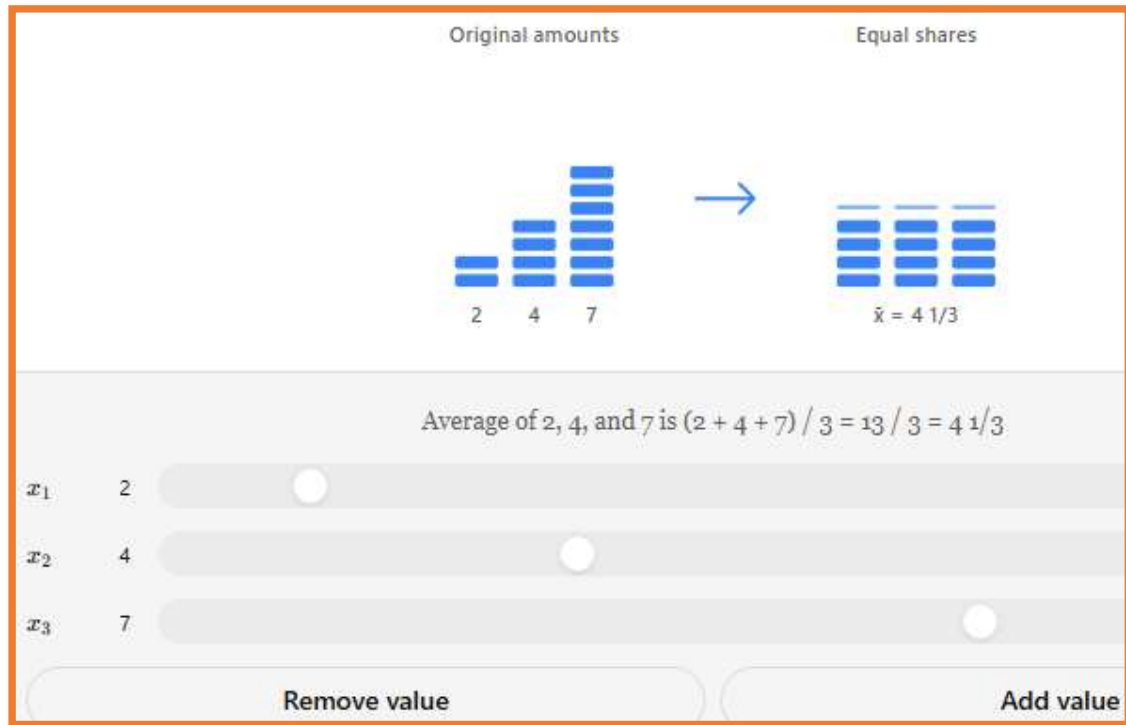
Sum of observations:

$$120 + 125 + 128 + 130 + 135 + 140 + 500 = 1278$$

Number of observations:

$$n = 7$$

$$\text{Mean} = \frac{1278}{7} = 182.57 \text{ ms}$$



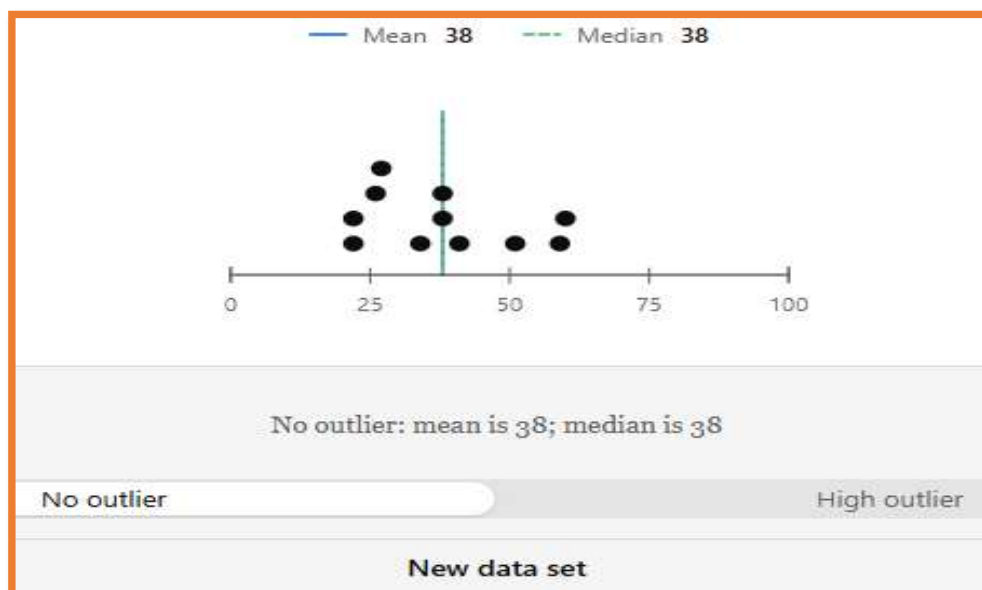
Median

Since there are **7 observations (odd number)**,

Median = 4th observation

$$\text{Median} = 130 \text{ ms}$$

(b) Which measure of central tendency is more appropriate? Justify your answer.



The **median** is the more appropriate measure of central tendency.

Justification:

- a) The value **500 ms** is an **outlier**, much larger than the other response times.
- b) The **mean (182.57 ms)** is heavily influenced by this extreme value, making it much higher than the typical response time.
- c) The **median (130 ms)** is not affected by the outlier and better represents the center of the dataset.

Median is the preferred measure because the dataset contains an outlier.

(c) Calculate the Range and Interquartile Range (IQR)

Range

$$\begin{aligned}\text{Range} &= \text{Maximum} - \text{Minimum} \\ &= 500 - 120 = 380 \text{ ms}\end{aligned}$$

$$\text{Range} = 380 \text{ ms}$$

Interquartile Range (IQR)

Ordered data:

120, 125, 128, 130, 135, 140, 500

Median = 130

Lower half:

120, 125, 128

$$Q_1 = 125$$

Upper half:

135, 140, 500

$$Q_3 = 140$$

Therefore,

$$\begin{aligned}\text{IQR} &= Q_3 - Q_1 \\ &= 140 - 125 = 15 \text{ ms}\end{aligned}$$

$$\text{IQR} = 15 \text{ ms}$$

4. A dataset contains the following observations: 12, 15, 18, 20, 22, 25, 28, 30, 35

a) Compute the mean, median, and mode.

b) Compute the variance and standard deviation.

c) Interpret the variability of the dataset based on the standard deviation.

Solution:

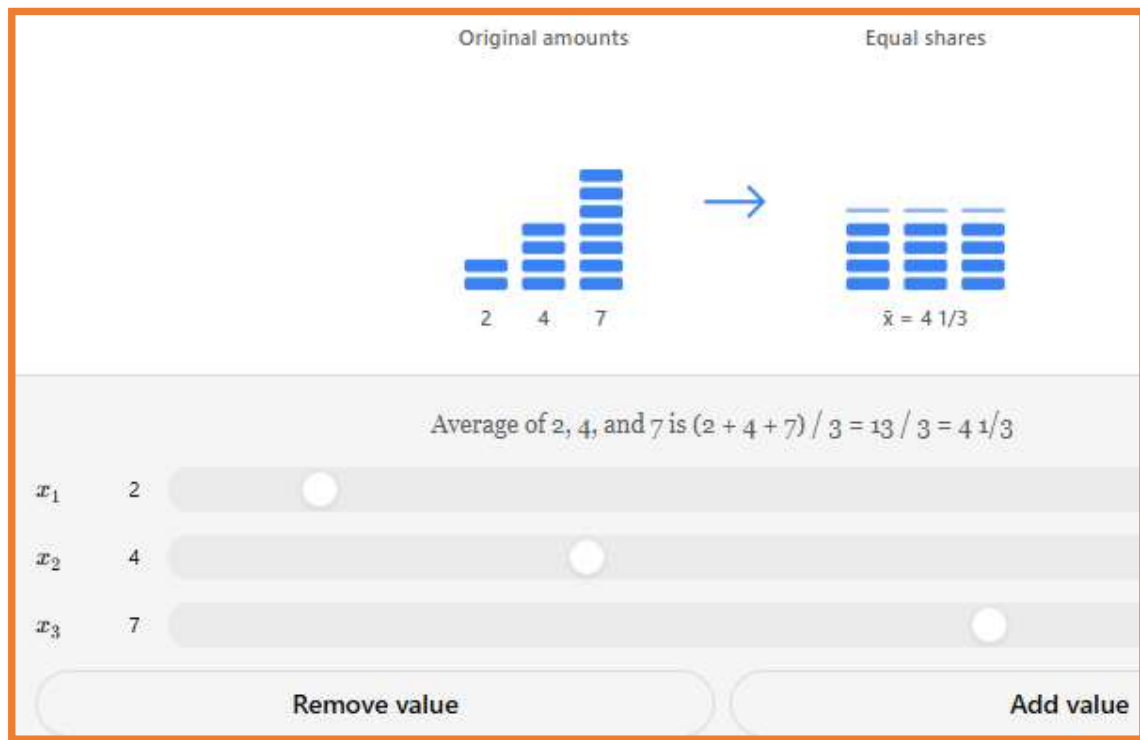
(a) Compute the Mean, Median, and Mode

Step 1: Arrange the data

The data are already in ascending order:

12, 15, 18, 20, 22, 25, 28, 30, 35

Mean



Sum of observations:

$$12 + 15 + 18 + 20 + 22 + 25 + 28 + 30 + 35 = 205$$

Number of observations:

$$n = 9$$

$$\bar{x} = \frac{205}{9} = 22.78$$

Median

Since there are **9 observations (odd number)**,

Median = 5th observation

$$\text{Median} = 22$$

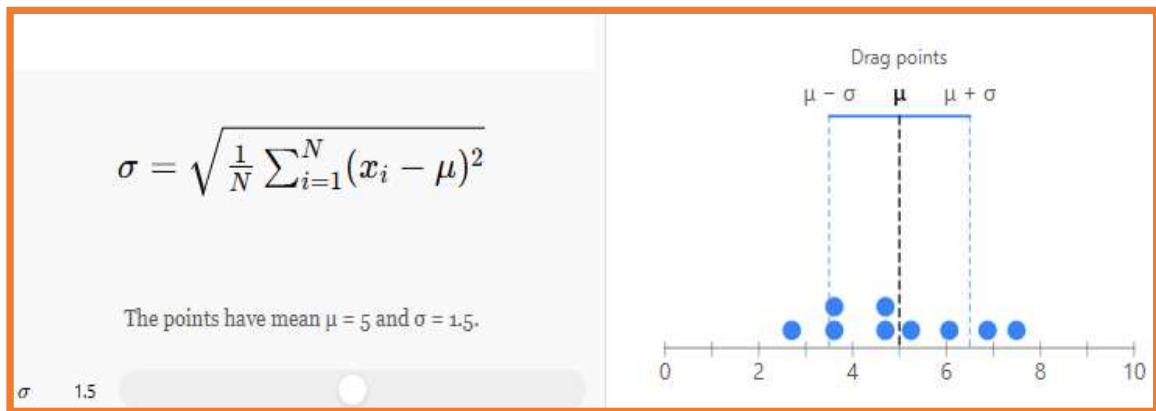
Mode

Each value appears only once.

Mode = No Mode

(b) Compute the Variance and Standard Deviation

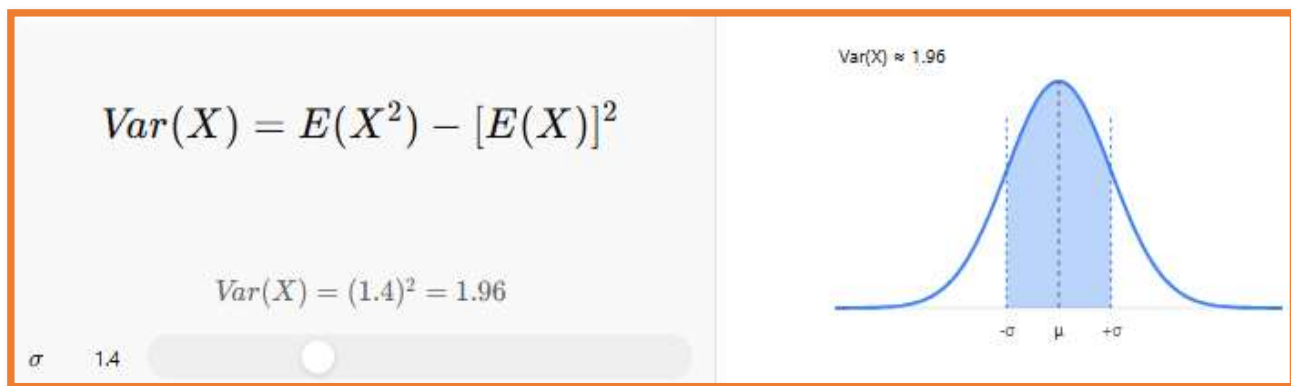
Calculate Squared Deviations



Mean = **22.78**

x	$x - \bar{x}$	$(x - \bar{x})^2$
12	-10.78	116.16
15	-7.78	60.53
18	-4.78	22.84
20	-2.78	7.73
22	-0.78	0.60
25	2.22	4.94
28	5.22	27.27
30	7.22	52.16
35	12.22	149.38
Total		441.61

Population Variance



$$= \frac{441.61}{9} = 49.07$$

$$\boxed{\text{Population Variance} = 49.07}$$

Population Standard Deviation

$$\sigma = \sqrt{49.07} = 7.01$$

$$\boxed{\text{Population Standard Deviation} = 7.01}$$

(Optional) Sample Variance and Standard Deviation

If the data are considered a **sample**,

$$s^2 = \frac{441.61}{8} = 55.20$$

$$s = \sqrt{55.20} = 7.43$$

$$\boxed{\text{Sample Variance} = 55.20}$$

$$\boxed{\text{Sample Standard Deviation} = 7.43}$$

(c) Interpretation of the Standard Deviation

The **population standard deviation is approximately 7.01**.

Interpretation:

- a) The standard deviation indicates that the observations typically vary by about **7 units** from the mean (**22.78**).
- b) Since the standard deviation is **moderate** relative to the mean, the data show a **moderate spread** around the average.
- c) There are **no extreme outliers**, and the observations are reasonably distributed around the mean.

$\boxed{\text{The dataset exhibits moderate variability, with most values lying within about } \pm 7 \text{ units of the mean.}}$

→ **Interpretation:** The dataset has **moderate variability**. Most observations are clustered around the mean (22.78), with a typical deviation of about **7 units**, indicating no unusually large spread or extreme outliers.

5. The marks scored by 12 students are: 40, 42, 45, 47, 48, 50, 52, 54, 56, 58, 60, 95

a) Calculate the mean and median.

b) Calculate the range and IQR.

c) Comment on the effect of the extreme value (95) on the measures of central tendency.

Solution:

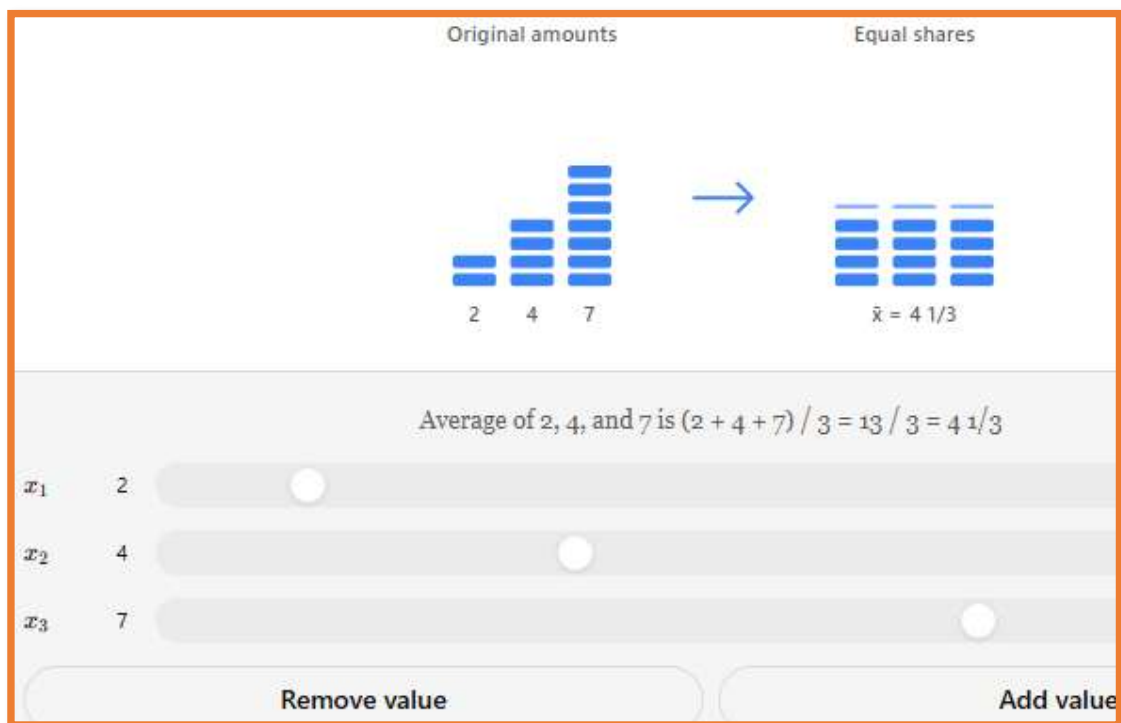
(a) Calculate the Mean and Median

Step 1: Arrange the data

The data are already in ascending order:

40, 42, 45, 47, 48, 50, 52, 54, 56, 58, 60, 95

Mean



Sum of marks:

$$40 + 42 + 45 + 47 + 48 + 50 + 52 + 54 + 56 + 58 + 60 + 95 = 647$$

Number of students:

$$n = 12$$

$$\text{Mean} = \frac{647}{12} = 53.92$$

Median

Since there are **12 observations (even number)**,

Median is the average of the **6th** and **7th** observations.

6th observation = **50**

7th observation = **52**

$$\text{Median} = \frac{50 + 52}{2} = 51$$

$$\text{Median} = 51$$

(b) Calculate the Range and Interquartile Range (IQR)

Range

$$\begin{aligned} \text{Range} &= \text{Maximum} - \text{Minimum} \\ &= 95 - 40 = 55 \end{aligned}$$

$$\text{Range} = 55$$

Interquartile Range (IQR)

Data:

40, 42, 45, 47, 48, 50, 52, 54, 56, 58, 60, 95

Lower half:

40, 42, 45, 47, 48, 50

$$Q_1 = \frac{45 + 47}{2} = 46$$

Upper half:

52, 54, 56, 58, 60, 95

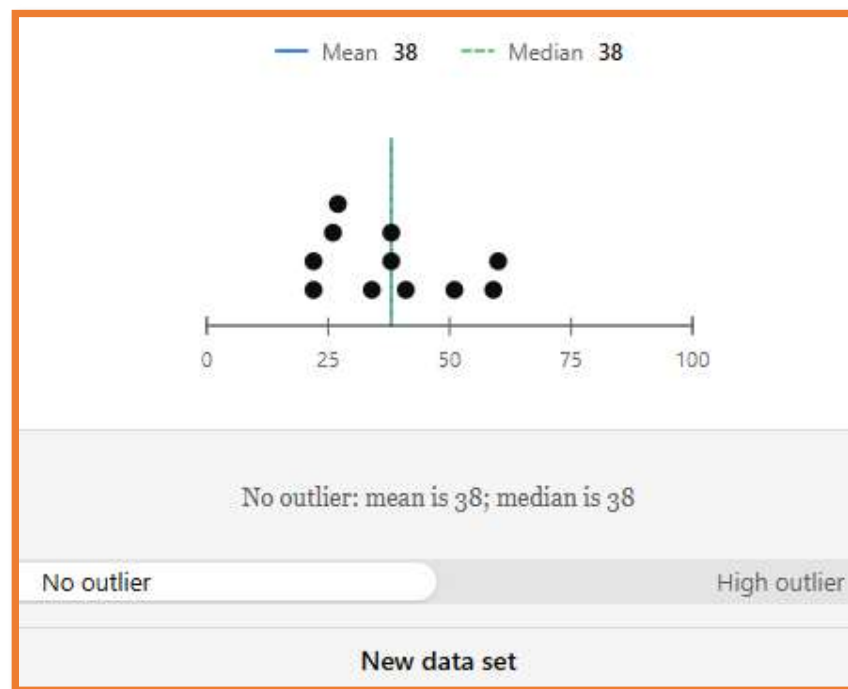
$$Q_3 = \frac{56 + 58}{2} = 57$$

Therefore,

$$\begin{aligned} \text{IQR} &= Q_3 - Q_1 \\ &= 57 - 46 = 11 \end{aligned}$$

$$\boxed{\text{IQR} = 11}$$

(c) Comment on the Effect of the Extreme Value (95) on the Measures of Central Tendency



The value **95** is an **extreme value (outlier)** because it is much larger than the other marks.

Effect on the Mean

- The **mean (53.92)** is increased by the extreme value.
- Without 95, the average would be much closer to the center of the remaining marks.

Effect on the Median

- a) The **median (51)** is almost unaffected by the extreme value because it depends only on the middle observations.
- b) It better represents the typical performance of the students.

Conclusion

The extreme value (95) has a significant effect on the mean but very little effect on the median.

Therefore, **the median is the more appropriate measure of central tendency** for this dataset.

→ Interpretation

- a) **95** is an outlier.
- b) It **increases the mean** considerably.
- c) The **median remains stable** and better reflects the typical marks of the students.
- d) Hence, **the median is the preferred measure of central tendency** for this dataset.

VIVA-VOCE Questions:

1. What is the difference between a population and a sample?

A **population** is the entire group under study, while a **sample** is a subset selected from the population.

2. Differentiate between a parameter and a statistic.

A **parameter** describes a population, whereas a **statistic** describes a sample.

3. When is the median preferred over the mean?

The **median** is preferred when the data contain **outliers** or are **skewed**.

4. What is the difference between variance and standard deviation?

Variance measures the average squared deviation from the mean, while **standard deviation** is the square root of the variance.

5. Why is the interquartile range (IQR) useful for skewed data?

The **IQR** measures the spread of the middle 50% of the data and is **not affected by outliers**.
